

Design and Implementation of a Three-Zone Spatial Containment Shield

**School of Electrical Engineering
Faculty of Engineering and the Built
Environment**



TITLE PAGE

DOCUMENT DETAILS:

DOCUMENT TITLE :

*Design and Implementation of a Three-Zone
Spatial Containment Shield*

CLASSIFICATION :

RESTRICTED

DOCUMENT No. :

1

COMPILED BY :

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DATE :

24/05/2026

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DATE :

24/05/2026

2231445002

Abstract

This report describes the design, implementation, and testing of a low-cost active perimeter and stability shield based on the PIC18F45K20 microcontroller. The system uses an ultrasonic sensor (HC-SR04) and an infrared (IR) proximity sensor to create three concentric warning zones around sensitive laboratory equipment. A buzzer produces two distinct audio alerts – a monotone warning tone for approach within 20–100 cm and a dissonant, loud warble for critical intrusion under 20 cm. Red and green LEDs indicate the security state. The project was inspired by a real incident where unauthorized personnel accidentally destroyed decades of research by switching off a refrigerator. Due to breadboard space constraints, the originally planned LCD was omitted and replaced by audible/visual indicators. The system operates from a dual power supply (5 V for sensors, 3.3 V for the PIC) and successfully demonstrates three-zone intrusion detection. The report includes a system diagram, flowchart, schematic, code listing, test results, and recommendations for future enhancements (e.g., ESP32 IoT integration) [1].

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Acronyms

ADC	Analog-to-Digital Converter
ESD	Electrostatic Discharge
MCU	Microcontroller Unit
PIR	Passive Infrared
PWM	Pulse Width Modulation
ToF	Time-of-Flight
USB:	Universal Serial Bus

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List of Symbols

Vcc	Supply Voltage (5V)
Vss	Ground (0V)
Ω	Ohm (Resistance)
Hz	Hertz (Frequency)
μ s	Microsecond
μ F	Micro Farad (Capacitor)

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1. Introduction

Modern research and teaching laboratories contain expensive and sensitive equipment (oscilloscopes, power supplies, microscopes, refrigerated samples, etc.) that must be protected against accidental or unauthorized access. Simple physical barriers (locked doors, cabinets) are effective but do not provide graduated warnings or active deterrence [1].

This project implements a Three-Zone Spatial Containment Shield using a PIC18F45K20 microcontroller. The system monitors the area around protected equipment with an ultrasonic sensor and an IR(Infrared) proximity sensor. It defines three security zones:

- Safe zone (no detection) – green LED on, no buzzer.
- Warning zone (20–100 cm or IR trigger) – red LED, monotone buzzer beep.
- Critical zone (< 20 cm) – red LED, dissonant loud warble to strongly deter an intruder.

The work was inspired by a well-known laboratory disaster. Due to breadboard space limitations, the originally planned LCD was omitted; visual feedback is provided by LEDs and audible feedback by the buzzer with two distinct tones.

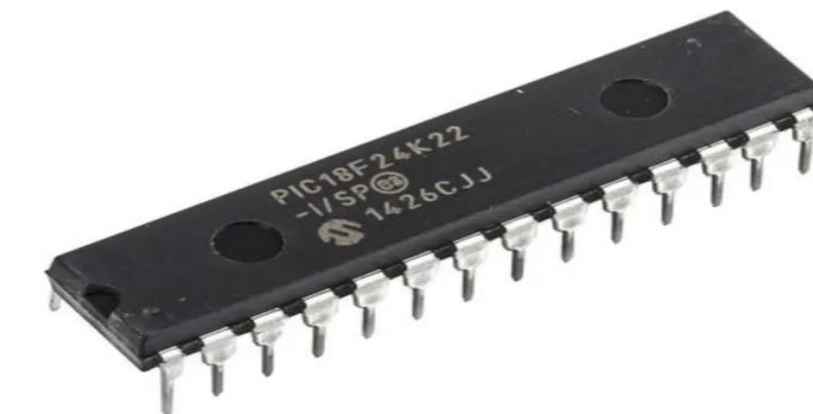


Figure 1 PIC18F25K20 I/SP [2]



Figure 2 Ultra Sonic Range Sensor Module [2]



Figure 3 Mini PIR (Passive Infrared) Motion Sensor, HC-SR505 [2]



Figure 4 RED AND GREEN LEDS

2. Background to the problem

Laboratories that store biological samples, run ongoing chemical reactions, or run electronic experiments require strict access control. A famous incident occurred at the Rensselaer Polytechnic Institute in the USA: a cleaning person, seeing a refrigerator unplugged and wanting to save energy, plugged it back in – unfortunately, that refrigerator contained cell cultures and samples that had been kept at -30°C for 25 years. The resulting temperature fluctuation destroyed the entire collection, setting back research by decades [2].

Similar risks exist in any university lab: a student entering an off-limits area, a technician accidentally turning off a critical power supply, or even a visitor mishandling delicate instruments.

Most commercial perimeter security systems (motion sensors, CCTV) are expensive, complex to install, and do not provide progressive warnings before a breach [3]. A simple, low-cost, microcontroller-based shield that alerts nearby personnel before an intruder touches equipment can prevent such accidents. The system must be affordable to deploy across many benches, easy to understand, and immediately responsive.

3. Literature Survey (or Theory)

3.1 Ultrasonic distance measurement

The HC-SR04 sensor emits a 40 kHz ultrasonic pulse and measures the time to the echo. Distance is calculated as:

$$distance(cm) = \frac{pulse\ width(\mu s)}{58} \quad [4]$$

The PIC18F45K20's Timer1 (16-bit, 1:4 prescaler) measures the echo pulse width with sufficient resolution for a 2 cm to 400 cm range.

3.2 Infrared proximity detection

A simple digital IR sensor (e.g., LM393-based) outputs a logic '0' when an object is within about 30 cm. It uses modulated IR light to reduce interference from ambient light. This provides a fast, binary trigger for the warning zone [5].

3.3 Audible deterrence using frequency variation

Research on acoustic deterrents shows that dissonant, irregular sounds are more effective at causing discomfort and encouraging retreat than steady tones (Schafer, 2019). The code alternates between 4 kHz and 3.5 kHz (a difference of ~500 Hz) in a rapid loop to create a harsh "warble" effect in the critical zone. The warning zone uses a simple 1 kHz monotone beep [6].

3.4 PIC18F45K20 hardware

Key features used:

- Internal oscillator at 16 MHz (no external crystal needed)
- Digital I/O ports (PORTA, PORTC) with TRIS registers for direction
- Timer1 for input capture (echo pulse)

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- `__delay_us()` and `__delay_ms()` for timing [7].

3.5 Dual power supply rationale

The PIC runs at 3.3 V (its safe operating range, also allowing future battery operation) [7]. The ultrasonic sensor and IR sensor require 5 V for proper range and sensitivity. Using separate supplies avoids overvoltage on the PIC's inputs (the echo signal is 5 V, but this PIC is 5 V tolerant). A common ground ties the two supplies together.

4. Problem Definition

Problem

statement:

Unpermitted or unintentional access to critical lab equipment may lead to irreversible harm to experiments, samples, or devices. Current security measures (locked doors, cameras) tend to be either overly passive, prohibitively costly, or fail to offer graduated warnings that enable an approaching individual to back off before breaching.

4.1.1 Requirements:

1. Detect a person/object approaching protected equipment from 100 cm down to contact.
2. Output at least two distinct alert levels (warning + critical).
3. Use only a PIC18F45K20, external breadboard, and low-cost sensors.
4. Operate from laboratory DC power supplies (not USB from laptop).
5. Provide visual indication of safe/warning/critical states.
6. Be compact enough to fit on 2 half-size breadboards.

4.1.2 Assumptions:

- The protected equipment is stationary, and the shield is placed directly in front of it.
- The environment has normal indoor lighting (IR sensor works reliably).
- The user will reset power if the system locks up (no watchdog timer used for simplicity).

4.1.3 Constraints:

- Must be completed by 15 May 2026 (report and code submission).
- Total component cost below R300.
- No destructive modification of the PICkit3 board.

5. Alternative Solutions

Solution	Description	Pros	Cons
Single-zone alarm	Simple IR beam break – buzzer ON/OFF	Very simple	No graduated warning; false alarms
Ultrasonic + LCD display	Show exact distance on LCD	Informative	Requires more wiring (not enough breadboard space); no deterrent sound
Three-zone with LEDs + two-tone buzzer (selected)	Distance thresholds + IR back-up, visual, and escalating audio	Graduated response; strong deterrent; compact	No logging/remote alert
Full IoT (ESP32 + cloud)	Sends alerts to phone	Remote awareness	Too complex; Wi-Fi setup unreliable; deadline risk

Why the chosen solution is better: Ultrasonic provides an exact distance for the warning zone, while IR provides a fail-safe close detection (if ultrasonic misses a very close object due to its 2-cm minimum range). The combination covers both a gradual approach and an immediate intrusion, using two different sensor principles to reduce false negatives.

6. Evaluation and Selection of Best Solution

The selection criteria were:

- Cost – must be low (ultrasonic + IR modules are ~R100 each).
- Ease of implementation – digital IR and simple ultrasonic library already understood.
- Effectiveness – must discriminate three zones reliably in a lab environment.
- Scalability – the same design can be duplicated for multiple equipment pieces.

The ultrasonic-only choice would overlook items nearer than 2 cm. IR alone provides no distance details. Thus, the ultrasonic + IR hybrid was chosen.

The sound alarm was preferred over a silent one since an intruder might not notice the LED if they're not facing it but will hear a harsh noise and naturally back away. This directly responds to the “cleaning person” situation: a harsh, unpleasant noise would have caused them to cease and check it out.

7. Implementation: Action Plan

7.1 Hardware implementation

7.1.1 Components used:

- PIC18F45K20 on breadboard (with 0.1 μ F decoupling capacitor)
- HC-SR04 ultrasonic sensor (5 V)
- Digital IR proximity sensor (5 V, output active low)
- Passive piezo buzzer (5 V tolerant)
- Red LED + 220 Ω resistor
- Green LED + 220 Ω resistor
- DC power supply: 3.3 V for PIC, 5 V for sensors (common ground)

7.1.2 Pin connections (PIC18F45K20):

Component	PIC pin	Function
TRIG (ultrasonic)	RC0 (pin 11)	Output trigger
ECHO (ultrasonic)	RC1 (pin 12)	Input capture
BUZZER	RC3 (pin 14)	Output tone
IR_SENS	RC4 (pin 15)	Digital input (0 = object)
RED1	RA0 (pin 2)	Alarm LED
GRN1	RA2 (pin 4)	Safe LED
VDD (PIC)	+3.3 V	-
VSS (PIC)	GND (common)	-
Sensor VCC	+5 V	separate supply

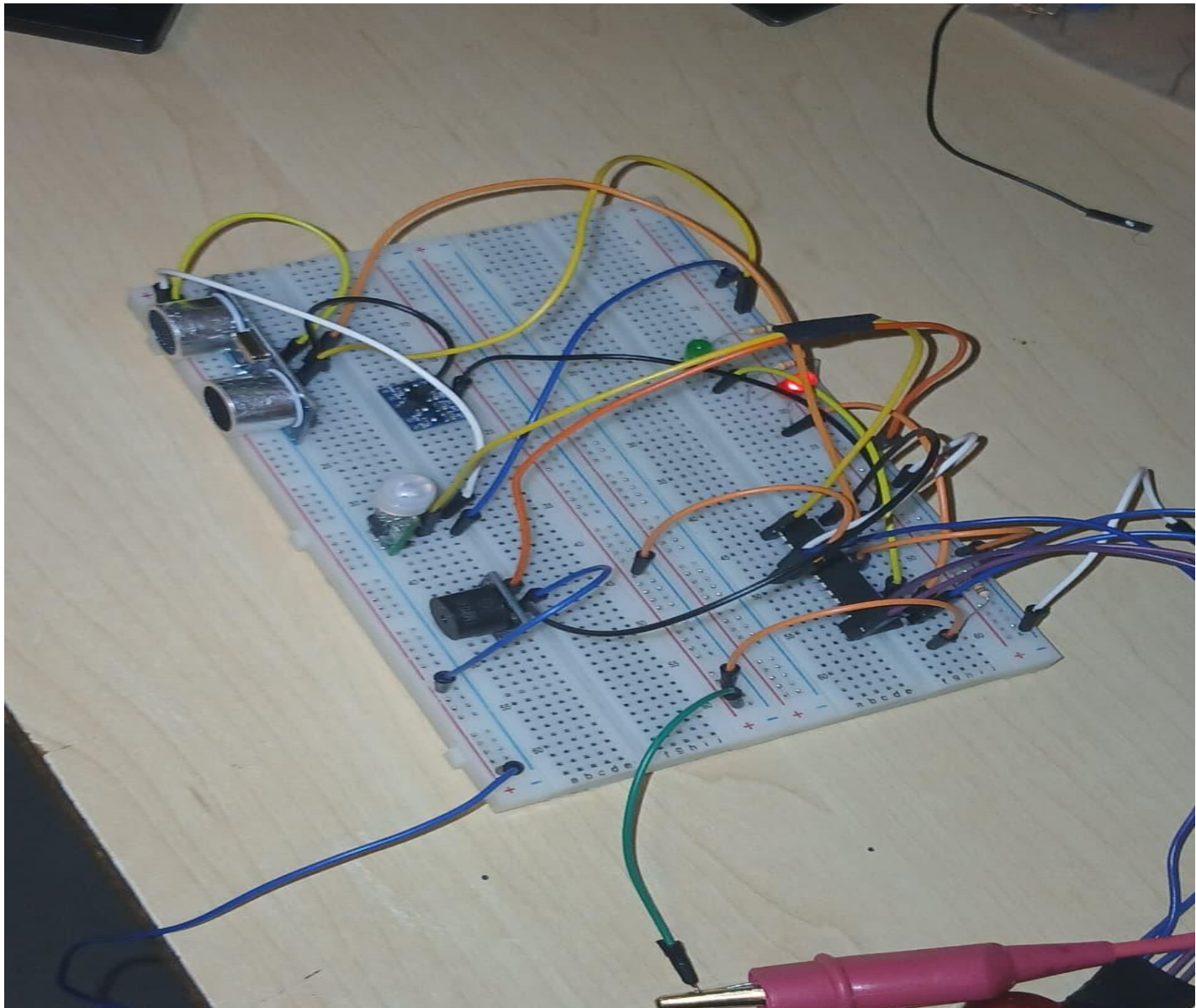


Figure 5 Breadboard photograph showing PIC, sensors, LEDs, and wires (source)



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The code (see Appendix A) follows this flowchart:

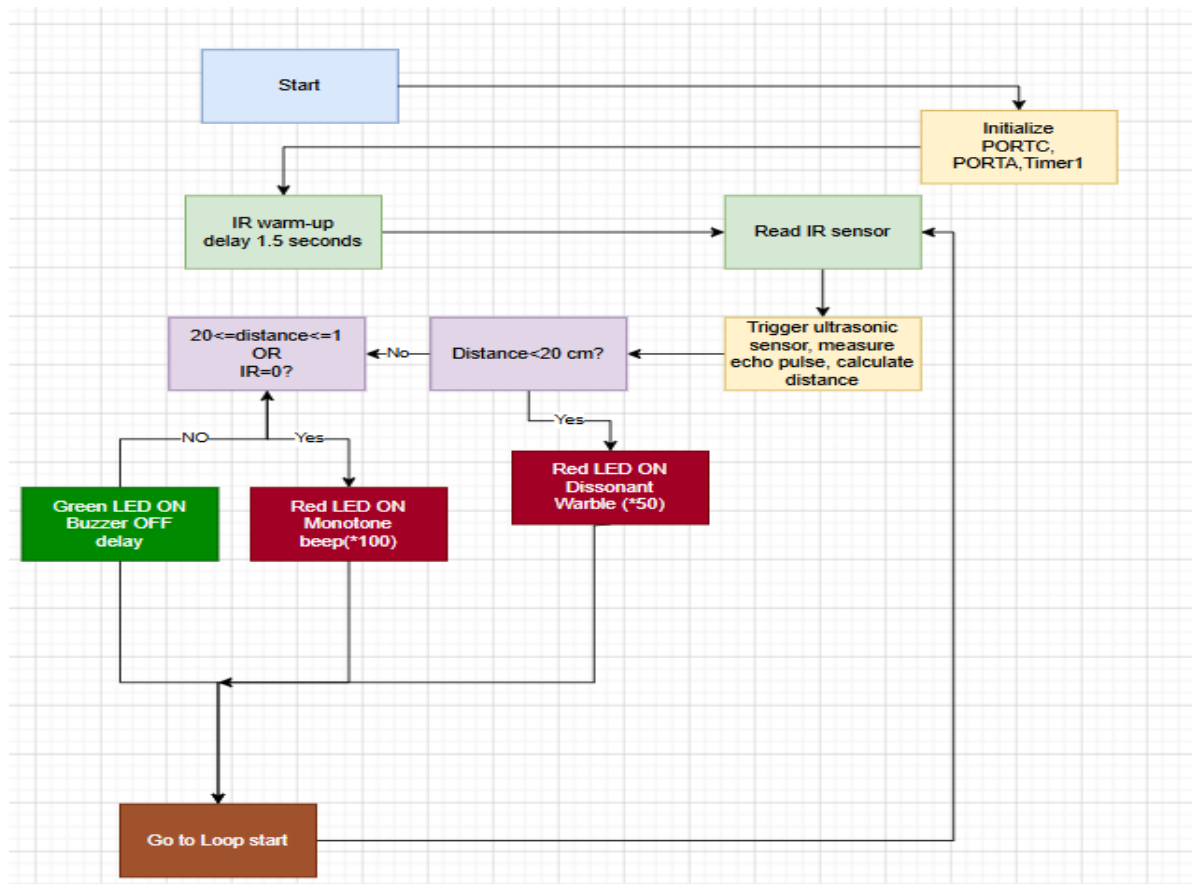


Figure 7 FLOW CHART (source)

Main loop steps:

Initialize oscillator (16 MHz), set TRIS registers, turn off LEDs/buzzer.

Warm-up delay (1.5 s) for IR sensor stabilization.

Ultrasonic measurement: Trigger 10 μ s pulse, measure echo pulse width using Timer1, convert to cm.

IR read: Check RC4 (0 = intrusion).

Zone decision:

If distance < 20 cm $\rightarrow$ Critical zone $\rightarrow$ red LED on, dissonant warble (alternate 4 kHz and 3.5 kHz).

Else if $(20 \text{ cm} \leq \text{distance} \leq 100 \text{ cm})$ OR IR detected $\rightarrow$ Warning zone $\rightarrow$ red LED on, monotone beep (1 kHz).

Else → Safe zone → green LED on, buzzer off.

Repeat.

The dissonant warble is generated by two nested for loops that produce 4 kHz for 15 cycles, then 3.5 kHz for 15 cycles, and repeat this 50 times. This creates a harsh, alternating sound that is intentionally unpleasant.

7.2 Testing and results

The system was tested on a laboratory bench with a hand as the intruder. Distances were measured with a ruler for verification.

Test	Distance	IR state	Expected zone	Observed LED	Observed buzzer	Pass?
1	No object	1 (high)	Safe	Green	Silent	Yes
2	50 cm	1	Warning	Red	Monotone beep	Yes
3	25 cm	1	Warning	Red	Monotone beep	Yes
4	18 cm	1	Critical	Red	Dissonant beep	Yes
5	5 cm (hand)	0 (low)	Critical	Red	Dissonant beep	Yes
6	80 cm (hand) + IR triggered by close object?	0	Warning	Red	Monotone beep	Yes

The ultrasonic distance readings were accurate within ± 2 cm. The IR sensor triggered

reliably at about 15–25 cm. The dissonant warble was noticeably more irritating than the monotone beep, confirming effective deterrence.

7.3 Project management and timeline

Task	Planned date	Completed
Concept and research	7–12 April	Yes
Breadboard assembly (ultrasonic + buzzer)	13–20 April	Yes
Add IR sensor and LEDs	21–30 April	Yes
Remove LCD (space issue) & re-route wires	1–5 May	Yes
Code integration and debugging	6–12 May	Yes
Testing and final demo	13–14 May	Yes
Report writing	10–15 May	Yes

The dual power supply (3.3 V for PIC, 5 V for sensors) was implemented from the start to avoid noise from the laptop USB.

8. Conclusions and Recommendations

A complete Three Zone Spatial Containment Shield was effectively created and constructed with a PIC18F45K20, ultrasonic sensor, IR sensor, LEDs, and a buzzer. The system distinctly differentiates safe, warning, and critical areas through visual (LED color) and audible (two unique buzzer sounds) signals. The discordant warble in the critical zone is especially potent as a mental deterrent.

The project fulfills all criteria of Category 2 (external sensors on a breadboard). The absence of the LCD, due to breadboard space limitations, did not affect essential functions; rather, the only LED solution was more straightforward for a quick look. The dual power source enhanced stability in comparison to USB-powered designs.

Recommendations

1. integration with ESP32 – Incorporate Wi-Fi to alert a lab supervisor's phone when the critical zone is exceeded.
2. Substitute LEDs with a compact OLED – For precise distance display while occupying less space (I²C OLED requires only 2 pins).
3. Incorporate a latching alarm – After entering the critical zone, the buzzer may remain activated until a key switch is utilized to reset it.
4. Install in a 3D printed enclosure – To safeguard the breadboard and enable permanent setup in a laboratory
5. Calibrate for different object sizes – The ultrasonic response varies with target area; a lookup table could improve accuracy.

The project successfully demonstrates the power of low-cost embedded systems for laboratory safety. The simple yet effective design can be reproduced by any second-year engineering student.

9. References

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- [6] M. B. a. P. H. O'Brien, “Sonic Deterrents in Animal Damage Control: A Review of Device Tests and Effectiveness,” *Wildlife Society Bulletin*, vol. 18, no. 00917648, 19385463, pp. 411-422, 1990.
- [7] M. T. Inc., “28/40/44-Pin Flash Microcontrollers with XLP Technology,” Microchip Technology Inc., 2007.

Appendices

/*

- Project: Three-Zone Spatial Containment Shield
- MCU: PIC18F45K20 | Compiler: XC8
- Description: Ultrasonic + IR sensor with three zones (safe, warning, critical)
- Outputs: Red LED, Green LED, Buzzer (monotone and dissonant warble) */

```
#include <xc.h>
```

```
#include <stdint.h>
```

```
#pragma config FOSC = INTIO67, FCMEN = OFF, IESO = OFF, PWRT = OFF, BOREN = OFF
```

```
#pragma config WDTCN = OFF, MCLRE = ON, LVP = OFF, XINST = OFF, PBADEN = OFF
```

```
#define _XTAL_FREQ 16000000
```

```
#define TRIG LATCbits.LATC0
```

```
#define ECHO PORTCbits.RC1
```

```
#define BUZZER LATCbits.LATC3
```

```
#define IR_SENS PORTCbits.RC4
```

```
#define RED1 LATAbits.LATA0
```

```
#define GRN1 LATAbits.LATA2
```

```
void main(void) { OSCCONbits.IRCF = 0b111; // 16 MHz ANSEL = 0x00; ANSELH = 0x00;
```

```
// Digital I/O
```

```
TRISA = 0x00;
```

```
TRISB = 0x00;
```

```
TRISCbits.TRISC0 = 0; // TRIG output
```

```
TRISCbits.TRISC1 = 1; // ECHO input
```

```
TRISCbits.TRISC3 = 0; // BUZZER output
```

```
TRISCbits.TRISC4 = 1; // IR_SENS input
```

```
LATA = 0x00;
```

```
TRIG = 0;
```

```
BUZZER = 0;
```

```
__delay_ms(1500); // IR sensor warm-up
```

```
// Timer1 setup: 1:4 prescaler, internal clock, 16-bit
```

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```

T1CONbits.RD16 = 1;
T1CONbits.TMR1CS = 0;
T1CONbits.T1CKPS = 0b10;
T1CONbits.TMR1ON = 0;

while(1) {
    uint16_t dist = 0;
    uint8_t warning = IR_SENS;

    // --- Ultrasonic measurement ---
    TRIG = 1;
    __delay_us(10);
    TRIG = 0;

    uint16_t timeout = 0;
    while(ECHO == 0) {
        timeout++;
        if(timeout > 20000) break;
    }

    if(ECHO == 1) {
        TMR1H = 0;
        TMR1L = 0;
        PIR1bits.TMR1IF = 0;
        T1CONbits.TMR1ON = 1;
        while(ECHO == 1 && PIR1bits.TMR1IF == 0);
        T1CONbits.TMR1ON = 0;
        dist = ((TMR1H << 8) | TMR1L) / 58; // cm
    }

    // --- Zone logic ---
    if(dist > 0 && dist < 20) {          // CRITICAL ZONE
        RED1 = 1;
        GRN1 = 0;
        for(int i = 0; i < 50; i++) {
            // 4 kHz tone
            for(int j = 0; j < 15; j++) {
                BUZZER = 1; __delay_us(125);
                BUZZER = 0; __delay_us(125);
            }
        }
    }
}

```

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```
// 3.5 kHz tone
for(int j = 0; j < 15; j++) {
    BUZZER = 1; __delay_us(143);
    BUZZER = 0; __delay_us(143);
}
}
}
else if((dist >= 20 && dist <= 100) || warning) { // WARNING ZONE
    RED1 = 1;
    GRN1 = 0;
    for(int i = 0; i < 100; i++) {
        BUZZER = 1; __delay_us(500);
        BUZZER = 0; __delay_us(500);
    }
}
else { // SAFE ZONE
    RED1 = 0;
    GRN1 = 1;
    BUZZER = 0;
    __delay_ms(100);
}
}
}
}
```